

ECE 341 Homework 2

The problem numbers listed below are from the Lathi textbook, 3rd edition.

Prob. Number	Course Outcome	Points
2.2-1 (b, d)	1.b	20
2.3-2	1.b	20
2.4-18 *	1.b	10
2.4-27 (b, f)	1.b	30
2.5-1	7.c	20
TOTAL:		100

*** Hint: You can use Table 2.1 of the textbook (Page 176) to find the convolution.**

1-b. Evaluate and classify signals to identify unstable systems.

7-c. Apply new knowledge to understand continuous and discrete systems.

3.1.1 a) Taking the individual values of each non-zero $x[n]$ shown in the the figure,

$$\sum_{n=-2}^0 \left(\frac{3}{2}n + 3\right)^2 + \sum_{n=-1}^2 \left(-\frac{3}{2}n + 3\right)^2 = 1^2 + 2^2 + 3^2 + 2^2 + 1^2 = 19.$$

b) Again, taking the individual values of each non-zero $x[n]$,

$$\sum_{n=-3}^3 \left(\frac{9}{3}n\right)^2 = (9 + 9) + (36 + 36) + (81 + 81) = 252.$$